



New Mansoura University
Faculty of Computer Science and
Engineering



Lecture 4

Computer Architecture & Organization

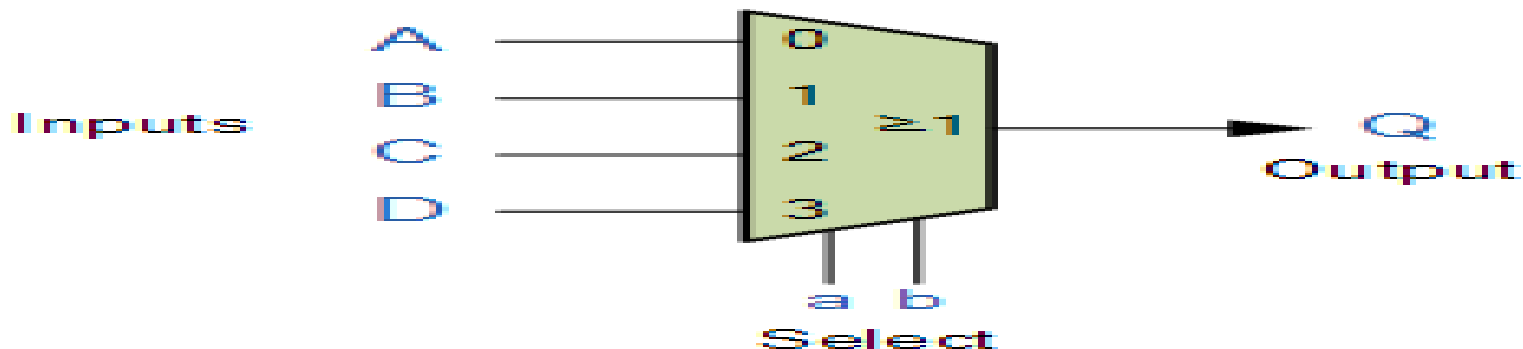
prepared by
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Bus Construction

- A bus consists of a set of common lines, one for each bit of a register, through which binary information is transferred one at a time.
- Control signals determine which register is selected by the bus during each particular register transfer.
- There are two ways to construct bus in computer
 - Multiplexers
 - Three state buffer

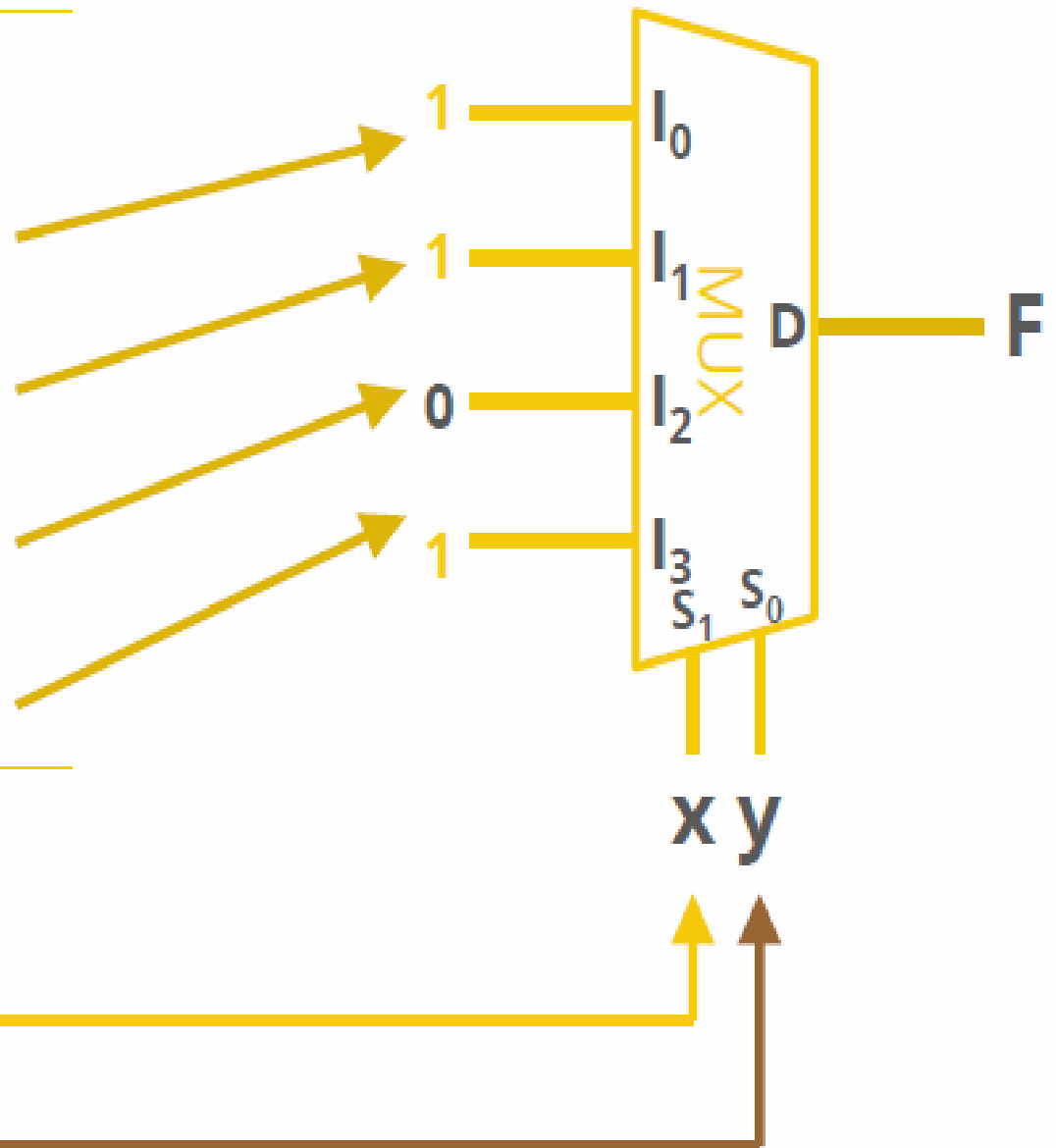
Multiplexer

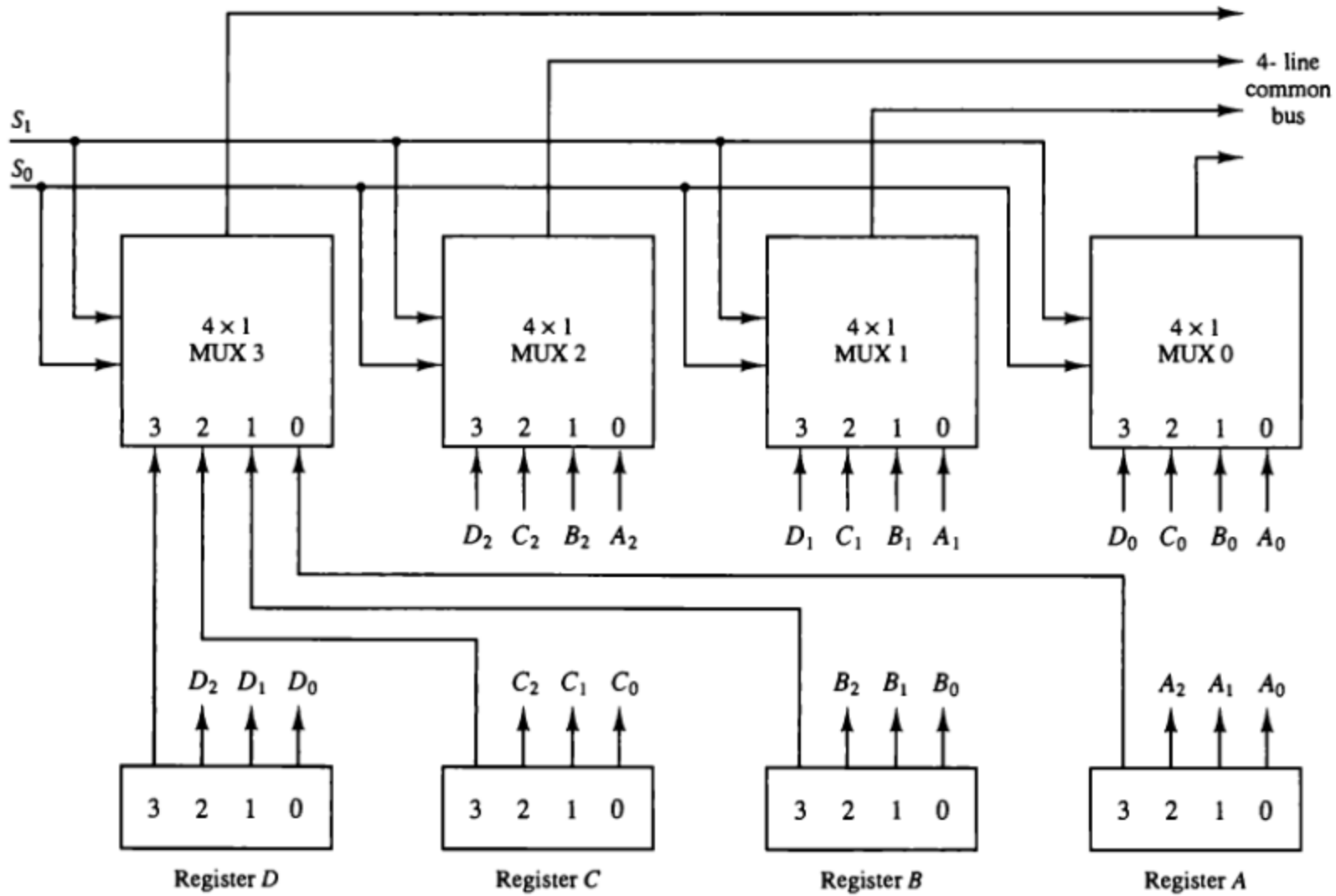
- In electronics, a multiplexer (or mux) is a device that selects one of several input signals and forwards the selected input into a single line.
- A multiplexer of 2^N inputs has n select lines, which are used to select which input line to send to the output.



Example

x	y	F
0	0	1
0	1	1
1	0	0
1	1	1





Exercise

A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers.

- a. How many selection inputs are there in each multiplexer?
- b. What size of multiplexers are needed?
- c. How many multiplexers are there in the bus?

- (a) 4 selection lines to select one of 16 registers.
- (b) 16×1 multiplexers.
- (c) 32 multiplexers, one for each bit of the registers.

Three-State Bus Buffers

- A bus system can be constructed with three-state gates instead of multiplexers.
- A three-state gate is a digital circuit that exhibits three states.
 - Two of the states are signals equivalent to logic 1 and 0 as in a conventional gate.
 - The third state is a high-impedance state.
- The high-impedance state behaves like an open circuit, which means that the output is disconnected and does not have a logic significance.

Figure 4-4 Graphic symbols for three-state buffer.



Design one bit bus for 4 register

- The control inputs to the buffers determine which of the four normal inputs will communicate with the bus line.
- No more than one buffer may be in the active state at any given time.
- One way to ensure that no more than one control input is active at any given time is to use a decoder.

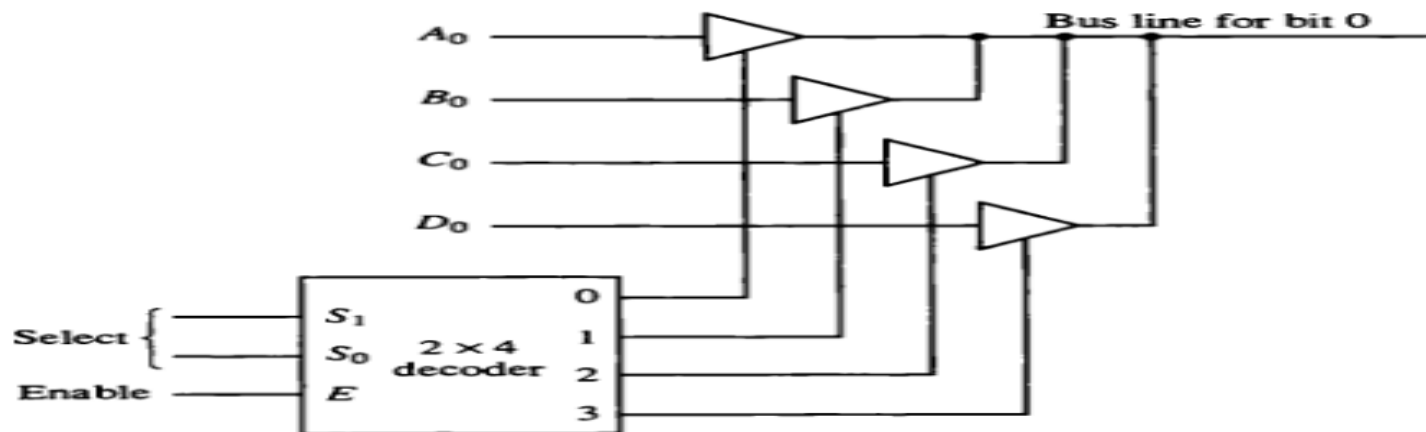
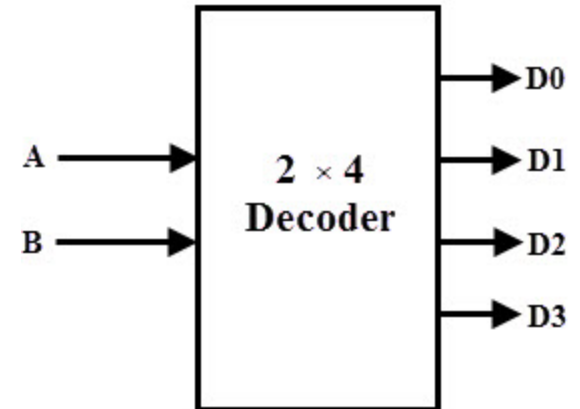


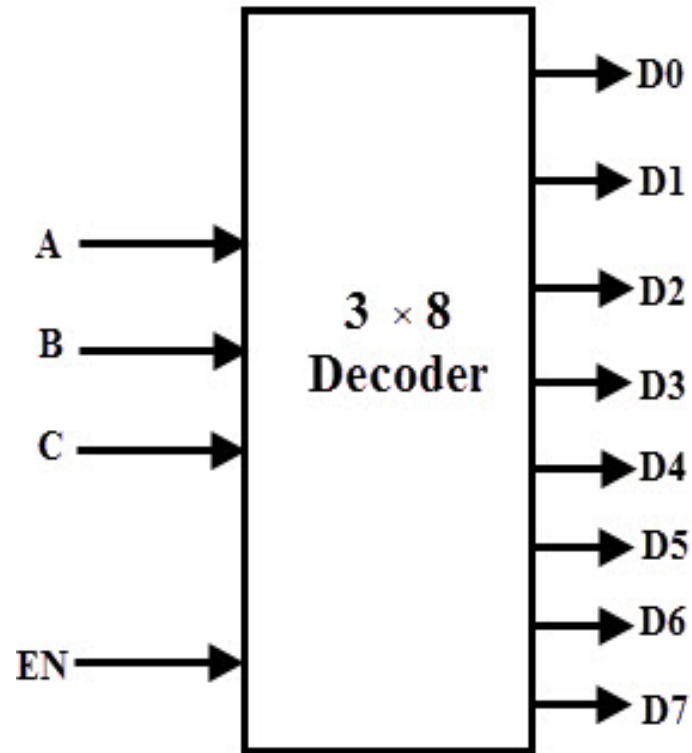
Figure 4-5 Bus line with three state-buffers.

Decoder

- In digital electronics, a **decoder** is a combinational logic circuit that **converts binary information from the n coded inputs to a maximum of 2^n unique outputs.**
- Example **2 to 4 decoder**



3 x 8 Decoder



Computer architecture according to memory architecture

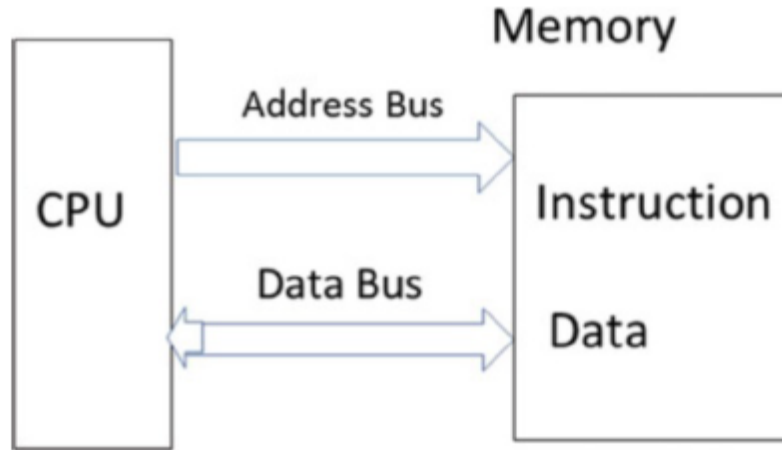
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- There are two types of Computer architecture, and they are von Neumann and Harvard architecture.
- The Von Neumann architecture is a computer design model where both data and instructions are stored in the same memory and accessed via a single bus.
 - Von Neumann uses the data bus to transfer data and instructions from the memory to the CPU.
- The Harvard architecture is a computer design model where instructions and data are stored in separate memory units and accessed through different buses.
 - Harvard Architecture Harvard architecture uses separate buses for instructions and data.

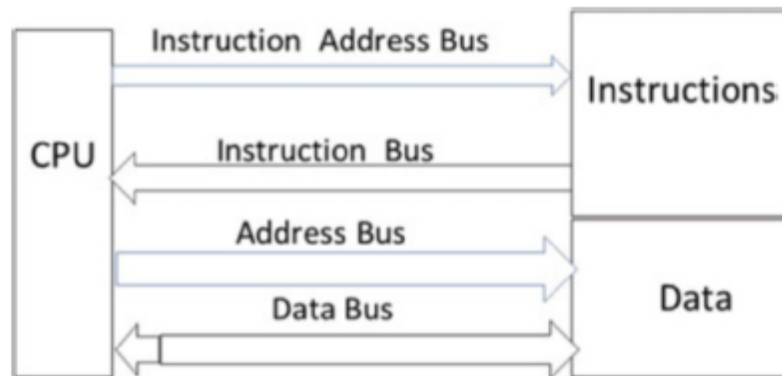
Computer architecture

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Von Neumann



Harvard



Harvard

- Two memories with two Buses allow parallel access to data access and instructions.
- Both memories can use different sizes.

Von Neumann

- Content of the memory is organised and all installed memory can be used.
- One bus is simpler for the control unit design
- Development of the Control Unit is cheaper and faster.
- Data and instruction is accessed in the same way.
- Computer with one bus is cheaper.
- Error in a program can rewrite instruction and crash program execution.

Harvard

- Development of a complicated Control Unit needs more time.
- Free data memory can't be used for instruction and vice-versa.
- Control unit for two buses is more complicated and more expensive.
- Program can't write

Von Neumann

- One Bus (for Data, instruction and devices) is a bottleneck.



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Thank You
For Your Attention!

Any Questions

